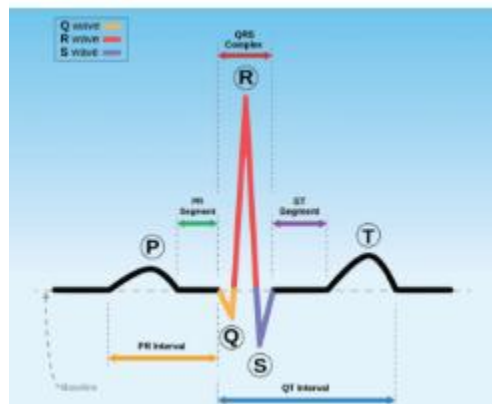


Theory:

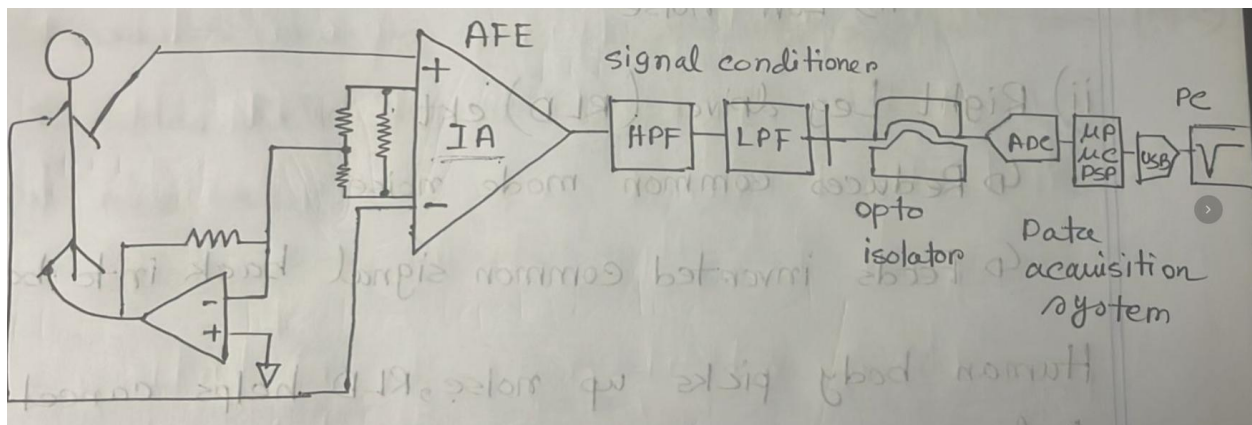
Electrocardiography (ECG) is a technique used to measure and record the electrical activity of the heart over time using electrodes placed on the surface of the body. The rhythmic contraction of heart is controlled by electrical impulses that originate within specialized cardiac cells. The ECG signal is essentially a graphical representation of those electrical events as they propagate through the heart.

Origin of the ECG signal lies in the process of depolarization and repolarization of the cardiac cells. Cardiac muscle cells maintain a resting potential across their membranes. When stimulated, ion channels open, causing a rapid influx of ions leading to the change of electrical potential. This process is called depolarization which triggers contraction of muscle fibers. After contraction, the cells return to their resting state through repolarization, where ions are restored to their original distribution. These electrical changes propagate in a coordinated manner across the heart, forming a wave of electrical activity. SA node of heart acts as a natural pacemaker, the generated impulses here spread through the atria, causing atrial depolarization and contraction. This wave reaches the AV node, where it encounters 0.1 second delay for the proper filling of ventricles. The impulse then faces ventricular depolarization stage, which results in ventricular contraction. Finally, the ventricles undergo repolarization, returning to their resting state in preparation for the next cycle. These electrical events are reflected in the characteristic ECG waveform, which consists of distinct components. The P-wave represents atrial depolarization and is usually a small, smooth, upward deflection. Following this is the QRS complex, which corresponds to ventricular depolarization. It is the most prominent part of the ECG waveform due to the larger muscle mass of the ventricles. The QRS complex consists of three components: the Q-wave (a small downward deflection), the R-wave (a large upward peak), and the S-wave (a downward deflection following the R-wave). After the QRS complex, the T-wave appears, representing ventricular repolarization. The T-wave is generally broader and smoother than the QRS complex. Atrial repolarization also occurs but is usually not visible because it is masked by the much larger QRS complex.



Leads in ECG provide the viewpoint of heart. ECG has 12 leads, among them 6 are limb leads and 6 are chest leads. The limb leads are bipolar leads that measure the voltage difference between two limbs. There are three standard limb leads: Lead I measures the potential difference between the left arm and right arm, Lead II between the left leg and right arm, and Lead III between the left leg and left arm. These three leads form a geometrical arrangement known as Einthoven's triangle, which helps in understanding the electrical axis of the heart. In addition to limb leads, there are augmented leads, which are unipolar leads derived from the limb electrodes. These include aVR (augmented voltage right arm), aVL (augmented voltage left arm), and aVF (augmented voltage foot). Each of these leads measures the electrical potential at one electrode relative to a reference formed by combining the other two electrodes. These leads provide additional angular perspectives of the heart's electrical activity. Finally, the chest leads, also known as precordial leads (V1 to V6), are placed directly on the chest at specific anatomical positions. These leads provide a horizontal or transverse view of the heart and are particularly useful for detecting abnormalities in specific regions of the heart muscle.

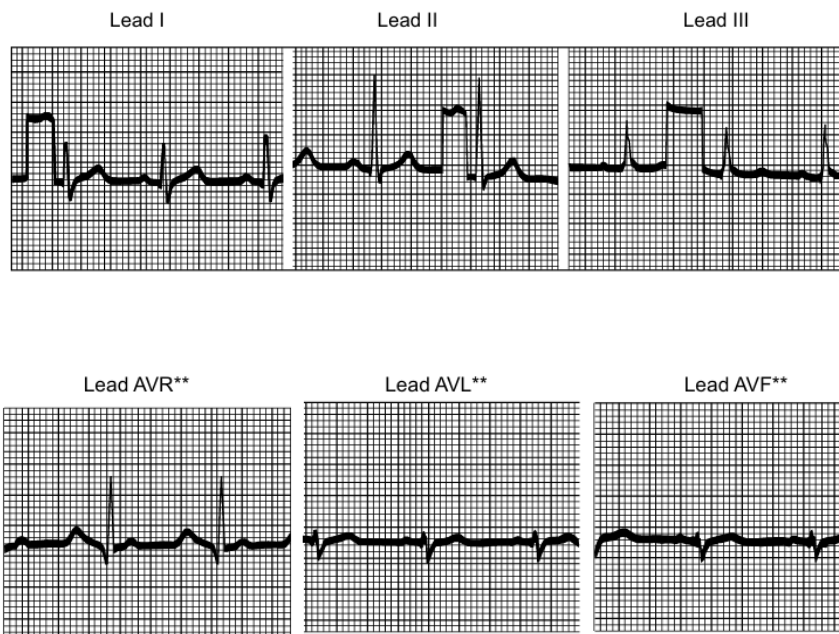
Circuit diagram:

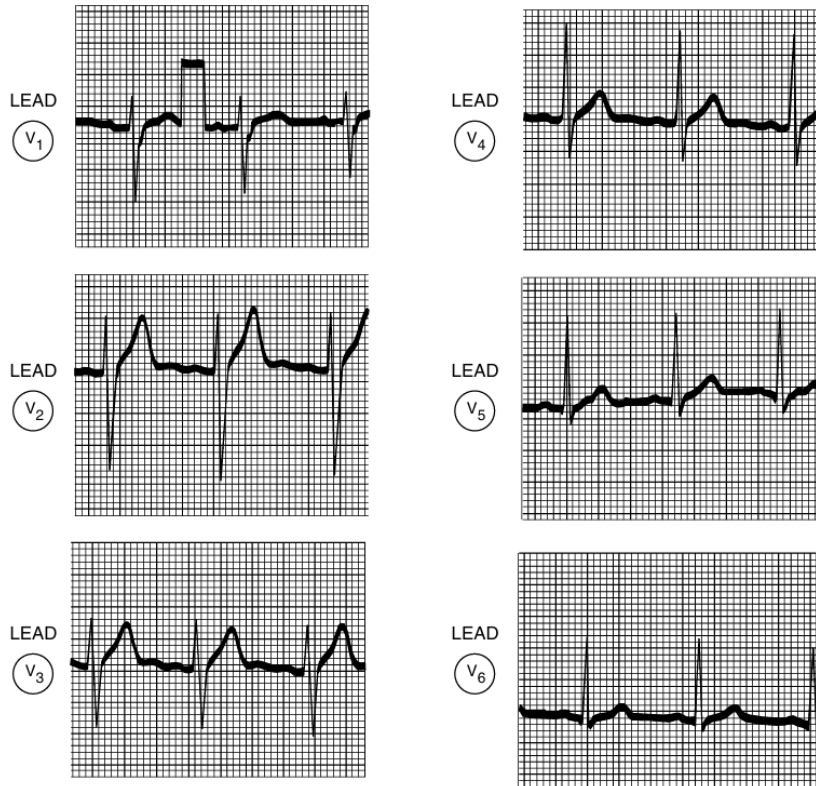


1. **Electrode and Lead Configuration**
They are placed on the patient's body to detect electrical signals, they convert ionic current in the body into electrical signals.
2. **Analog Front End**
This part includes instrumentation amplifier and an RLD circuit. The amplifier amplifies the difference between two electrode signals, rejects common-mode noise and ensures low noise. RLD circuit is used to reduce common mode noise and improves the overall signal quality.
3. **Signal Conditioning unit**
This block processes the amplified signal to remove noise and improve quality. It includes HPF to remove low frequency noise, LPF to remove high frequency noise, Notch filter to remove power line interference.

4. Opto-Isolator
Provides electrical isolation between patient and electronic system
5. Data Acquisition Unit
Converts analog ECG signal into digital form using Analog-to-Digital Converter (ADC).
6. PC Interface
Received digital ECG data and display the waveform on screen.

Results





Discussion:

1. ECG signals need amplification and filtering. Our heart generates small electrical signals which are in the range of 1 mV. Such small amplitude signals get distorted by noise easily and can not be analyzed properly. By the amplification and filtering of these signals, signal amplitude is increased and cleared, so that the generated waveform can be analyzed accurately.
2. Common sources of noise in ECG signal

- i) Power line interference: Typically occurs at 50/60 Hz depending on the region.
 - ii) Motion artifacts: Occur when the patient moves or when the electrodes shift position on the skin.
 - iii) Electrode contact noise: Poor contact between the electrode and skin, dry electrodes, or improper application of conductive gel can introduce noise.
 - iv) Baseline drift: Slow variation of the ECG baseline caused by respiration, body movement, or changes in electrode impedance.
3. Instrumentation amplifier: This is the first stage of the signal conditioning unit. This unit amplifies the difference between two input signals while rejecting common mode noise. The high input impedance ensures that it does not load the electrodes, preserving signal integrity. In this unit the RLD circuit feeds the body with inverted signals so that the picked up noises by body get eliminated.
- Filters help in removing the unwanted frequencies. High pass filters remove baseline drift and low frequency noise, LPFs remove high frequency interference. Notch filter specifically removes the power line interference.
- Gain stage ensures that the signal amplitude is sufficiently increased for further processing. Without amplification, the signal may remain too weak.
- Opto-isolation unit ensures patient safety by electrically isolating the patient from the rest of the system. This prevents any possibility of electrical shock.